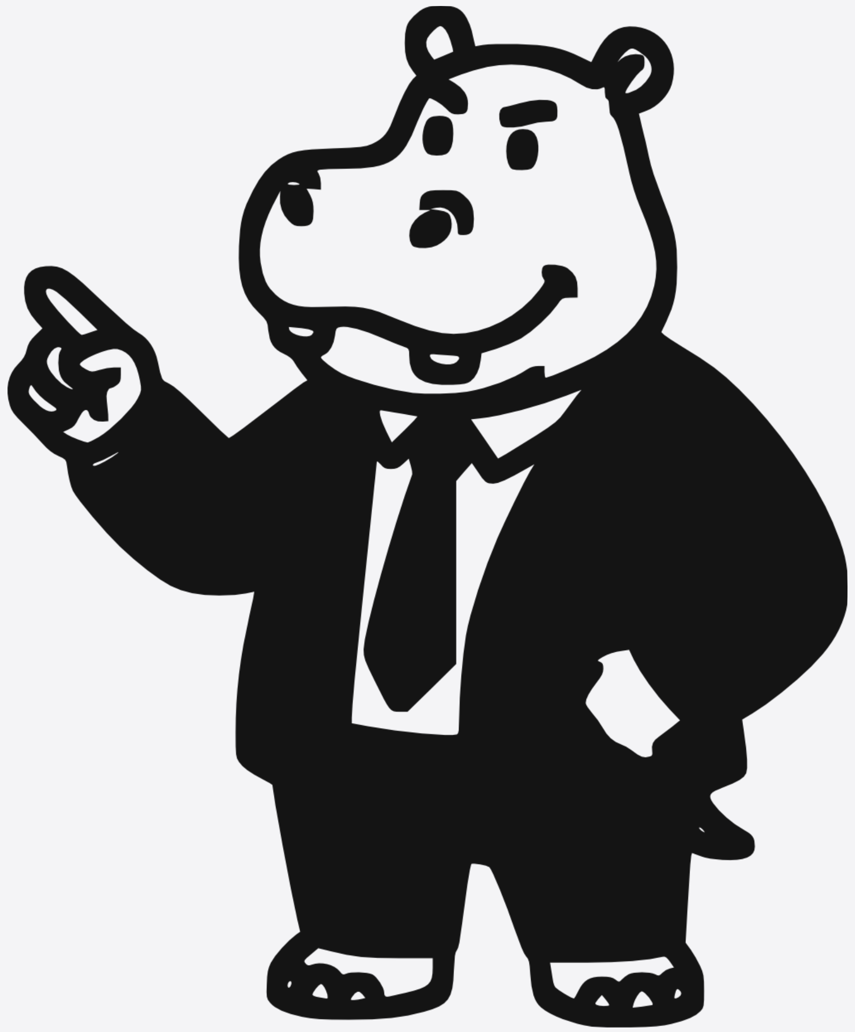
OPENING QUESTION

Has your design recommendation ever been **overruled?**

Because someone more senior had a different opinion.
That pattern has a name:
the **HiPPO effect** (McAfee & Brynjolfsson, 2012).



THE GAP IN PRACTICE

Fifteen professionals. Zero systematic frameworks.

Across fifteen expert interviews: UX designers, technical leads, quality engineers, project managers, not one described a systematic process for choosing an interface type.

> "Decisions here are not grounded in evidence. It is much more authoritarian than evidence-based."
>
> UX/Digital Designer
> 15 yrs experience

> "Basically, what prevents me from presenting a structured proposal to the client is the client themselves."
>
> Delivery Project Manager
> 8 yrs experience

> "You grab the template you like the most. There is no way to say which interface type suits the project better."
>
> Senior Fullstack Developer
> 10 yrs experience

THE GAP

Good tools for how to design. Nothing for what to design.

Every design methodology assumes you have already chosen what kind of interface to build.

None of them help you make that choice.

Refining a chosen interface Improving it, fitting it to users, testing and iterating	Exists
Choosing the interface type Screen · voice · wearable · AI · spatial: which one?	Missing

WHAT HAPPENS WITHOUT A FRAMEWORK

Right technology.
Wrong interface type.

Google Glass

Google Glass
2013

VALUES MISMATCH

Rejected on privacy, not technology.

Humane AI Pin

Humane AI Pin
2024

LATENCY + REDUNDANCY

5 to 10 second response delays.

Rabbit R1

Rabbit R1
2024

VALUE PROPOSITION

"Why not just use my phone?"

Fire Phone

Fire Phone
2014

TASK MISMATCH

$170M writedown. No task required.

Clippy

Clippy
2001

INTERACTION INITIATION

Interrupted users. They demanded removal.

Millions in combined documented losses across just five cases. Only the numbers companies disclosed.

IT IS NOT AN ISOLATED PATTERN

The failure is not technical.
It is structural.

95%

of generative AI pilots fail to deliver measurable impact.

MIT NANDA · 2025

80%

of AI projects never reach real use, twice conventional IT.

RAND Corporation · 2024

84%

of those failures trace to leadership decisions, not engineering.

RAND Corporation · 2024

> The bottleneck is the decision. Not the technology.

Master's Thesis Defense · 2026

• BEYOND INTUITION:

A Designer's Framework for Interface Type Selection

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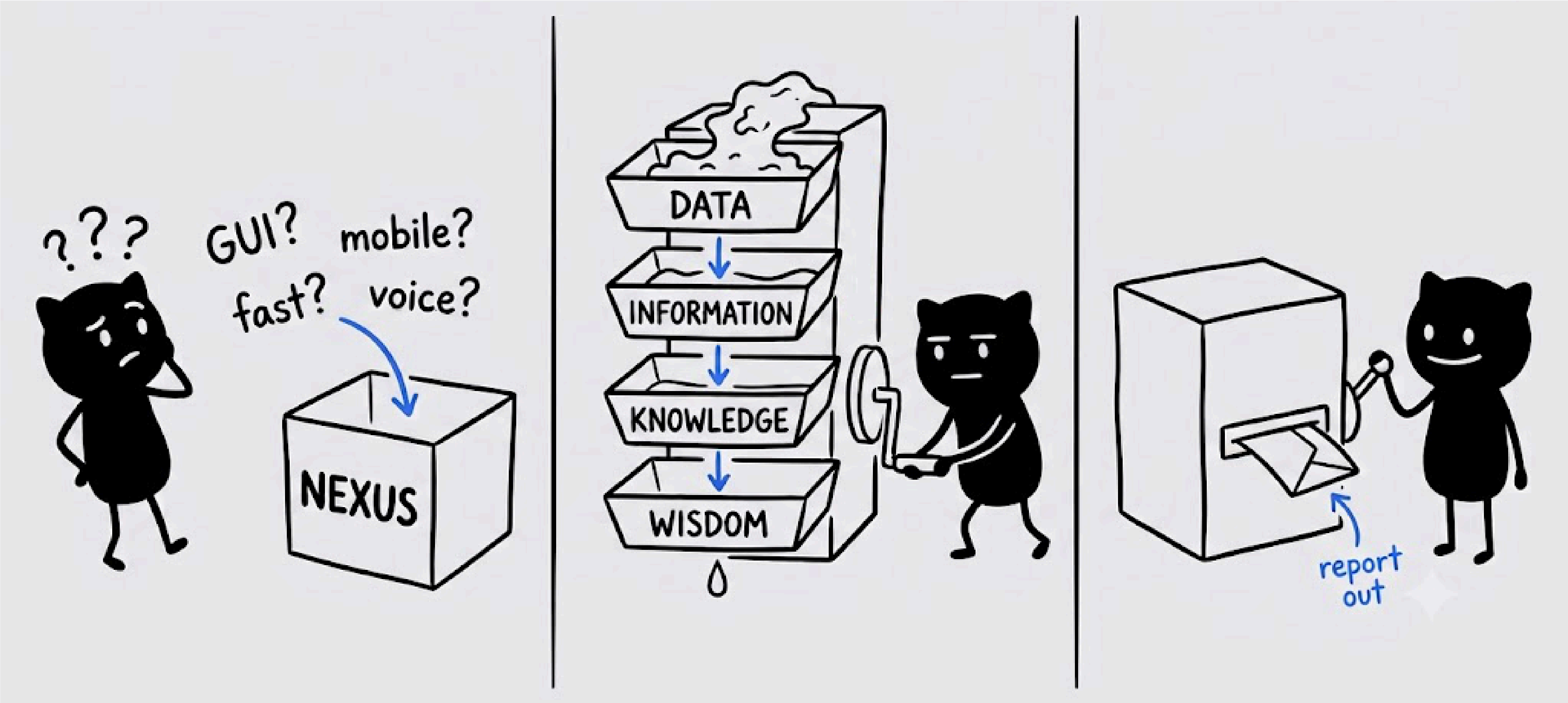
WHAT I BUILT

Give it a decision.
Get back evidence.



Your decision | NEXUS | A structured report

THE FRAMEWORK

A structured conversation, in four stages.

The decision that happens before the first mockup. Made explicit, documented, and defensible.

01 — **Values**
Rank what the organisation stands for. That ranking is the primary filter.

02 — **Context**
Fifteen questions: users, frequency, environment, consequence of failure.

03 — **Patterns**
Does this resemble any of the five documented failures? The framework flags it.

04 — **Report**
Not a verdict. Alignments, tensions, precedents. You read it. You decide.

15 questions, four DIKW layers.

Each question maps to a layer of the DIKW hierarchy established by Ackoff in 1989. The framework cannot be short-circuited: values are always evaluated before any technical question.

WISDOM	KNOWLEDGE	INFORMATION	DATA
Values ranking	**Context factors**	**Risk factors**	**Constraints**
What does your organisation actually stand for? Primary filter. Evaluated before any technical question.	Task complexity, frequency, predictability, context of use, information type. Who are the users and how do they work?	Error consequence, control preference, user demographics, exploration mode. What happens if something goes wrong?	Product type, existing solutions, interaction initiation, geography. Hard limits that constrain viable interface types.

RESEARCH METHOD

Three phases.
Three kinds of evidence.

PHASE 01 CONSTRUCTION

Built from the literature

Literature confirmed the gap. Interviews built the values taxonomy.

15 practitioners interviewed UX designers · technical leads · quality engineers · project managers

PHASE 02 RETROSPECTIVE

Tested against history

Applied to five failures, pre-launch information only.

5 cases pre-launch data only

PHASE 03 PROOF OF CONCEPT

Validated with practitioners

Built and validated with the same fifteen practitioners.

15 participants tool session

VALIDATION

Did it see what went **wrong?**

Applied to each product using only pre-launch public information.

PRODUCT	WHAT WENT WRONG	CAUGHT?
Google Glass	Privacy violation in public spaces	Yes
Humane AI Pin	Privacy + tech not ready for context	Yes
Rabbit R1	Problem users had already solved	Yes
Fire Phone	Innovation mismatched with phone use	Yes
Clippy	Automation that removed user control	Yes

82%

of documented failure factors identified across all five cases.

Target 70 % · Landis & Koch (1977)

70%

agreement with designer judgment

4.3/5

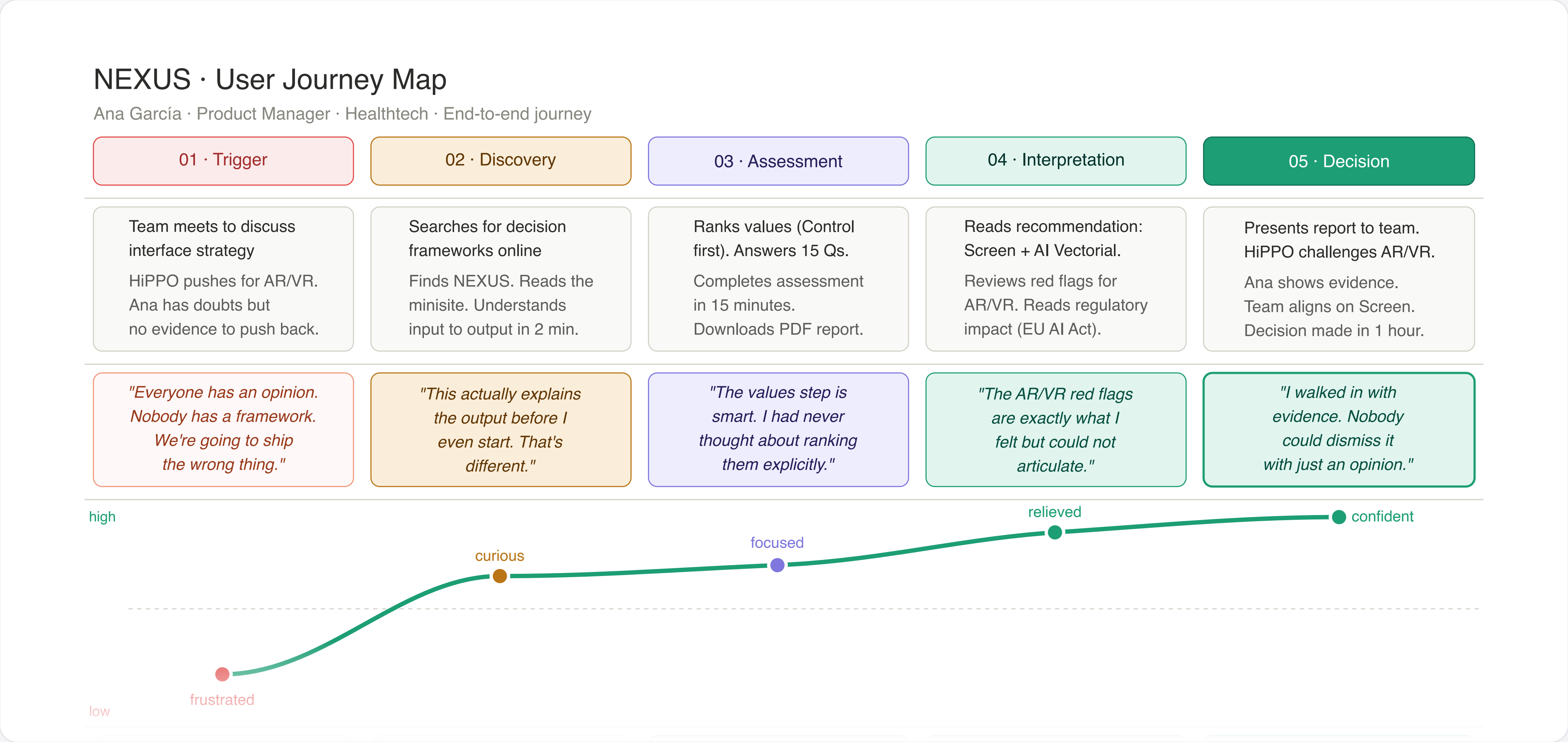
ease of use · 12 of 15 rated 4 or 5

WHO USES IT · UNDER WHAT CONDITIONS

From HiPPO pressure to evidence-backed decision.

Ana, a product manager in healthtech, walks into a meeting where AR/VR is being pushed without evidence. This is what happens next.

Emotion goes from frustrated to confident. The HiPPO is still in the room. But now so is the evidence.



THE TOOL

NEXUS in the browser.
Live demo.

- nexus-flame-delta.vercel.app/demo

Four contributions.
One open platform.

FOR THE FIELD

A replicable method

First application of DIKW to interface type selection. Extensible to new types, contexts, and failures not yet documented.

FOR PRACTICE

Evidence to defend good decisions

A structured argument for designers. Makes judgment visible, documented, and harder to override by opinion alone, including the HiPPO's.

FOR THE COMMUNITY

Infrastructure that outlives the defense

MIT-licensed. Open to all. The research does not end here.

AS A POSITION

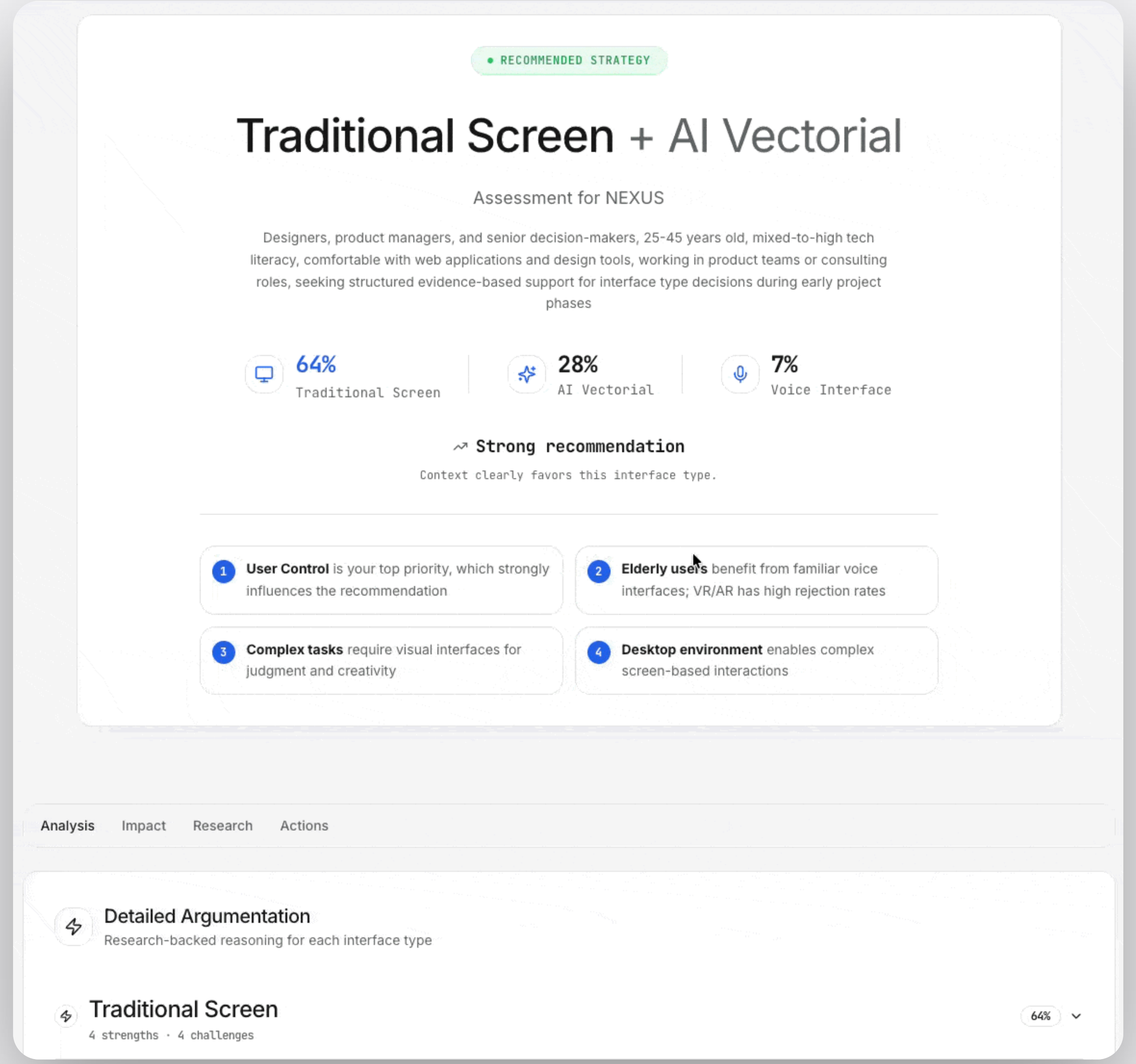
"It informs. It does not decide."

Structure in service of human judgment, not instead of it.

Under a minute.
Everything you just heard, in motion.

END · QUESTIONS WELCOME

Thank you.



Why DIKW and not Cynefin, AHP, or TAM?

Requirement	Design Thinking	TAM	MCDA	DIKW
Operates before type selection	No	No	Partial	Yes
Accommodates values without quantification	Yes	No	No	Yes
Enables cross-type comparison	No	Yes	Yes	Yes
Incorporates documented failure patterns	No	No	No	Yes
Accessible to non-specialist practitioners	Yes	No	No	Yes

Why these five cases and not others?

01
Commercial impact
Min. €100M investment or significant market withdrawal.

02
Documentation quality
Multiple independent sources: press, company comms, academic papers, user feedback.

03
Temporal coverage
1997–2024 across five interface types: automation, spatial, AI, screen, voice.

04
Interface type diversity
One case per category. Prevents skew from repeated failures in one type.

Are the patterns actually generalizable?

Recurrent across the corpus, not universal. That distinction matters.

WHAT THE CORPUS SUPPORTS

Five cases, 27 years, five interface types. All failures traceable to the same six pattern categories. Sufficient to structure a pre-selection checklist.

WHAT IT DOES NOT CLAIM

All cases are Western and high-budget. Other contexts may surface different patterns. The framework is designed to expand through community contribution.

> It structures the questions that make failure less invisible.

How does the scoring algorithm work?

01
Values filter first
Types contradicting top-ranked values are flagged before scoring. Hard filter, not a weight.

02
Context scoring
15 questions map to each type's strengths. Responses generate a compatibility profile per type.

03
Pattern matching
Failure triggers activate red flags with severity: blocking, critical, or warning.

04
Report generation
Alignments and tensions, not scores. The designer sees the reasoning, not just the result.

BACKUP · TECHNICAL

What if two interface types **score the same?**

A tie is not a failure. It is information.

WHAT IT MEANS

The context doesn't strongly favour either type. Both are viable given current inputs.

WHAT THE REPORT SHOWS

Both types side by side: red flags and alignments compared directly. Tension profiles, not scores.

WHAT THE DESIGNER DOES

Decides based on specific risks. The "informs, not decides" principle in practice.

How does NEXUS evolve after this defense?

Year 1 — **Single maintainer**
Author maintains quality. Community PRs accepted via structured review.

Year 2 — **Community steering**
Contributor council governs. Domain-specific forks enabled.

Year 3 — **Distributed ownership**
Full community control. Plugin architecture for swappable AI providers.

MIT license means anyone can use, modify, and extend it. No permission required, no payment needed.

Is this a master's thesis or a doctoral dissertation?

Deliberately bounded. One primary contribution, three subordinate.

PRIMARY — METHODOLOGICAL

Replicable retrospective validation and values-first filtering. The field can build on this independently of the platform.

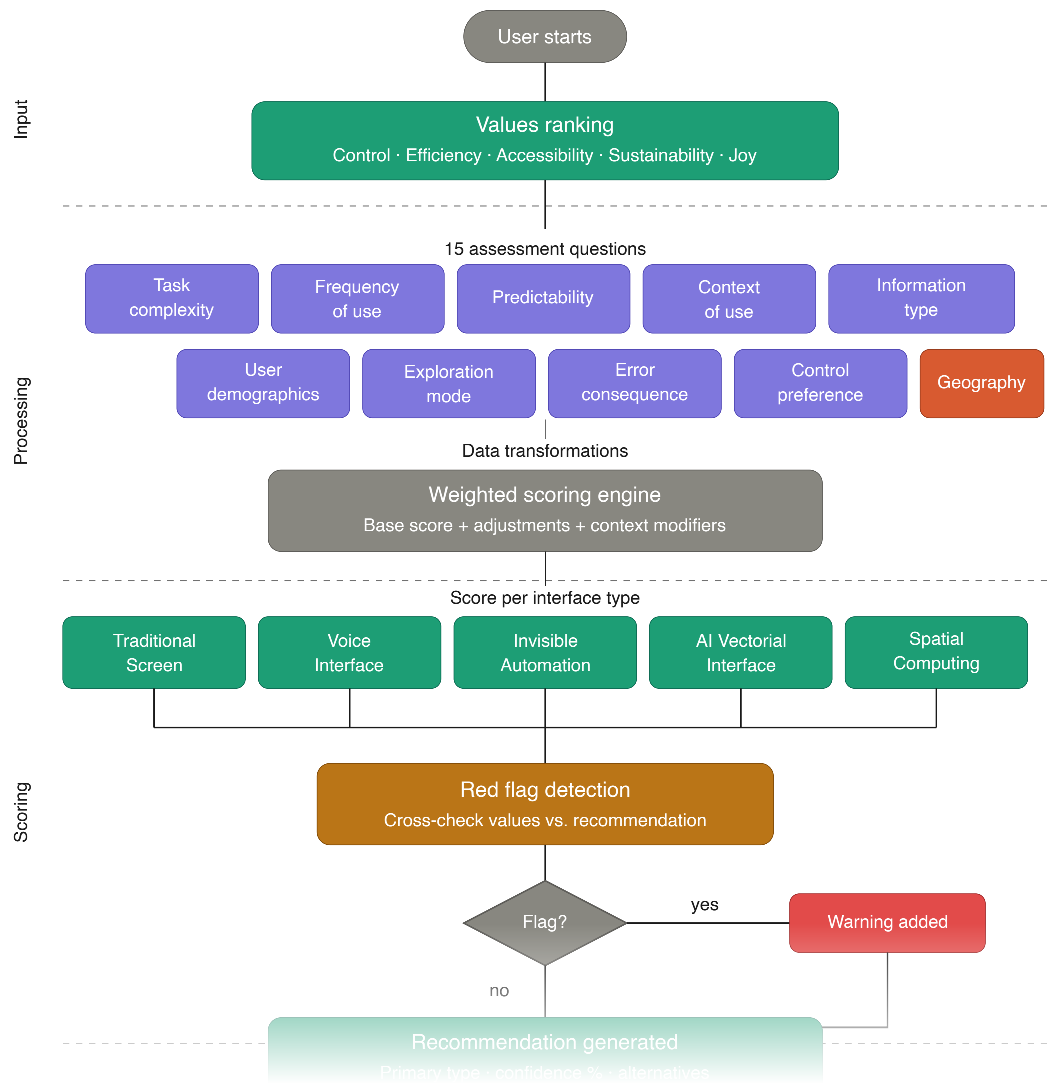
SUBORDINATE CONTRIBUTIONS

Theoretical: first DIKW application. Practical: NEXUS as proof of concept. Infrastructural: open-source governance model.

Field validation across organizations is the explicit next step, it is not a gap in this work.

Input to output.
Every step transparent.

From values ranking through scoring to recommendation. Each layer builds on the previous. The designer sees the reasoning behind every flag.



- Interface types and outputs
- Assessment factors (15 questions)
- Decision logic and red flag detection
- Processing nodes